

3. Three cholesterol lowering drugs are Lipitor (*L*), Zocor (*Z*), and Crestor (*C*). In a trial, 30 individuals with high cholesterol were randomly assigned, 10 to each group and the following sample means and sample variances were recorded, where the data consisted of reduction in cholesterol.

	Sample size J	Mean $\bar{X}$	Variance s^2
<i>L</i>	10	35	9
<i>Z</i>	10	25	9
<i>C</i>	10	50	16

Assume that the observations consisting of reduced cholesterol readings are independent, normally distributed with unknown means and unknown variance σ^2 . Determine, using the Tukey method, with $\alpha = .05$, which means differ significantly from each other.

4. A modification has been made to the process for producing a certain type of "time-zero" film (film that begins to develop as soon as a picture is taken). Because the modification involves extra cost, it will be incorporated only if sample data strongly indicates that the modification has decreased true average developing time by more than 1 second. Assume that the developing-time distributions differ only with respect to location if at all. Test at level .05, on the accompanying data, the appropriate hypothesis.

Original Process 8.6 5.1 4.5 5.4 6.3 6.6 5.7 8.5

Modified Process 5.5 4.0 3.8 6.0 5.8 4.9 7.0 5.7

Note: Do *not* assume normality. I.e., a *t*-test is not appropriate.